WORKSHEET 5.3

Increasing and Decreasing Intervals

Unit 5: Analysing Functions

Key ideas

- f is **increasing** on an interval where $f'(x) > 0$ and **decreasing** where $f'(x) < 0$.
- Find the critical points, then test the sign of f' on each interval between them.
- A factor that is always positive (a square, an exponential) does not affect the sign.
- The justification is a statement about the sign of f' on the whole interval.

Worked examples

Example 1 · A cubic

Find where $f(x) = 2x^3 - 3x^2 - 12x$ is increasing and decreasing.

Solution. $f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$, with signs $+, -, +$.

Increasing on $(-\infty, -1)$ and $(2, \infty)$; decreasing on $(-1, 2)$.

Example 2 · A logarithm

Find where $f(x) = x - 2\ln x$, $x > 0$, is increasing and decreasing.

Solution. $f'(x) = 1 - \dfrac{2}{x} = \dfrac{x-2}{x}$.

Decreasing on $(0, 2)$; increasing on $(2, \infty)$.

Example 3 · From the derivative

A function has $f'(x) = (x-1)(x+2)^2$. Where is f increasing?

Solution. $(x+2)^2 \geq 0$, so the sign of f' is the sign of $x - 1$.

For $x > 1$.

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Practice

CORE

1. Find where $f(x) = x^2 - 6x$ is decreasing.

2. Show that $f(x) = -x^3$ is never increasing.

3. Find where $f(x) = e^x - x$ is increasing.

4. A function has $f'(x) = x(x-4)$. On what intervals is f increasing?

EXAM LEVEL

5. Find where $f(x) = 3x^4 - 8x^3$ is decreasing.

6. Show that $f(x) = x + \cos x$ is never decreasing.

7. Find where $f(x) = x + \dfrac{3}{x}$ is decreasing.

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8. Find where $f(x) = x^2 - 2\ln x$, $x > 0$, is increasing.

Question 9 · Free-response practice

Let $f(x) = (x-3)^2 e^x$.

(a) Find $f'(x)$ in factored form.

(b) Find the intervals on which f is increasing and the interval on which f is decreasing. Justify your answer.

(c) Without computing, explain why $f(2) < f(1)$.

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Worked solutions

1. $f'(x) = 2x - 6 < 0$ for $x < 3$.
2. $f'(x) = -3x^2 \leq 0$ for all x.
3. $f'(x) = e^x - 1 > 0$ for $x > 0$.
4. $(-\infty, 0)$ and $(4, \infty)$.
5. $f'(x) = 12x^3 - 24x^2 = 12x^2(x-2) < 0$ for $x < 2$ $(x \neq 0)$: decreasing on $(-\infty, 2)$.
6. $f'(x) = 1 - \sin x \geq 0$ for all x.
7. $f'(x) = 1 - \dfrac{3}{x^2} < 0$ when $x^2 < 3$: on $\left(-\sqrt{3}, 0\right)$ and $\left(0, \sqrt{3}\right)$.
8. $f'(x) = 2x - \dfrac{2}{x} = \dfrac{2(x^2-1)}{x} > 0$ for $x > 1$.
9. (a) $2(x-3)e^x + (x-3)^2 e^x = (x-3)(x-1)e^x$.
   (b) $f' > 0$ for $x < 1$ and $x > 3$: increasing on $(-\infty, 1)$ and $(3, \infty)$. $f' < 0$ for $1 < x < 3$: decreasing on $(1, 3)$.
   (c) f is decreasing on $[1, 3]$ because $f'(x) < 0$ on $(1, 3)$, and $1 < 2$.

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